

112590024 邱紹澹 作業系統作業 HW3

7.17

- 4 個 goroutine
- 每個產生 100,000 個點

```
kelen@host:~/文件/GitHub/OSHW/HW3/OSHW_3_individual_112590024/7_17$ go run mulThread.go
Total Points : 400000
Circle Points: 313819
Estimated Pi : 3.138190
kelen@host:~/文件/GitHub/OSHW/HW3/OSHW_3_individual_112590024/7_17$ |
```

7_17.png

7.14

程式碼加入 mutex

```
/* Thread-safe wrappers */
static int ts_allocate_pid(void)
{
    pthread_mutex_lock(&pid_mutex);
    int pid = -1;
    if (g_initialized) {
        size_t idx;
        if (find_first_free(&idx)) { bit_set(idx); pid = MIN_PID + (int)idx; }
    }
    pthread_mutex_unlock(&pid_mutex);
    return pid;
}
```

mutex.png

```
kelen@laptop:~/文件/GitHub/OSHW/HW2/OSHW_2_indiv
$ gcc pid_ex.c
kelen@laptop:~/文件/GitHub/OSHW/HW2/OSHW_2_indiv
$ ./a.out
=== PID Manager - Multithreaded Test ===
PID range   : [300, 5000] (4701 slots)
Threads      : 100
Max sleep    : 5 s

[thread  0] PID 300 acquired → sleeping 5 s
[thread  1] PID 301 acquired → sleeping 4 s
[thread  3] PID 302 acquired → sleeping 1 s
[thread  2] PID 303 acquired → sleeping 3 s
[thread  4] PID 304 acquired → sleeping 2 s
[thread  5] PID 305 acquired → sleeping 3 s
[thread  6] PID 306 acquired → sleeping 1 s
[thread  7] PID 307 acquired → sleeping 2 s
[thread  8] PID 308 acquired → sleeping 5 s
[thread  9] PID 309 acquired → sleeping 3 s
[thread 11] PID 310 acquired → sleeping 4 s
[thread 12] PID 311 acquired → sleeping 5 s
[thread 10] PID 312 acquired → sleeping 3 s
[thread 13] PID 313 acquired → sleeping 2 s
[thread 14] PID 314 acquired → sleeping 4 s
```

7_14_1.png

```
[thread 32] PID 328 released
[thread 41] PID 341 released
[thread 55] PID 355 released
[thread 57] PID 357 released
[thread 60] PID 360 released
[thread 63] PID 363 released
[thread 64] PID 364 released
[thread 65] PID 365 released
[thread 66] PID 366 released
[thread 69] PID 369 released
[thread 74] PID 374 released
[thread 83] PID 383 released
[thread 84] PID 384 released
[thread 85] PID 385 released
[thread 86] PID 386 released
[thread 88] PID 388 released
[thread 93] PID 393 released
[thread 94] PID 394 released

=== Results ===
Successful allocations : 100
Failed allocations    : 0
Total threads        : 100
Bitmap clean: all acquired PIDs were properly released.
```

7_14_2.png

8.32

讓用戶可以輸入北方或南方各幾個農夫

```

kelen@host:~/文件/GitHub/OSHW/HW3/OSHW_3_individual_112590024/8_32$ ./po
Enter number of northbound farmers: 5
Enter number of southbound farmers: 3

=== Bridge Simulation Start ===

[Northbound 0] waiting to cross...
[Northbound 0] on the bridge, crossing...
[Northbound 1] waiting to cross...
[Northbound 2] waiting to cross...
[Northbound 3] waiting to cross...
[Northbound 4] waiting to cross...
[Southbound 0] waiting to cross...
[Southbound 1] waiting to cross...
[Southbound 2] waiting to cross...
[Northbound 0] crossed the bridge.
[Northbound 1] on the bridge, crossing...
[Northbound 1] crossed the bridge.
[Northbound 2] on the bridge, crossing...
[Northbound 2] crossed the bridge.
[Northbound 3] on the bridge, crossing...
[Northbound 3] crossed the bridge.
[Northbound 4] on the bridge, crossing...
[Northbound 4] crossed the bridge.
[Southbound 0] on the bridge, crossing...
[Southbound 0] crossed the bridge.
[Southbound 1] on the bridge, crossing...
[Southbound 1] crossed the bridge.
[Southbound 2] on the bridge, crossing...
[Southbound 2] crossed the bridge.

=== All farmers have crossed the bridge ===

```

8_32.png

9.28

```

kelen@host:~/文件/GitHub/OSHW/HW3/OSHW_3_individual_112590024/9_28$ gcc addresses.c -o addresses
kelen@host:~/文件/GitHub/OSHW/HW3/OSHW_3_individual_112590024/9_28$ ./addresses 19986
The address 19986 contains:
page number=4
offset=3602
kelen@host:~/文件/GitHub/OSHW/HW3/OSHW_3_individual_112590024/9_28$

```

9_28.png